Thermal radiation and color

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use peak wavelength as one piece of evidence about a thermal spectrum.

SUCCESS CRITERIA - I can...

1. Relate higher temperature to shorter peak wavelength.
2. Use kelvin in Wien's law.
3. Distinguish spectral peak from perceived color.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An object moves 0.3 m toward a fixed plane mirror. Find image movement and the change in object-image separation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\lambda_{max}\, T = 2.9\times10^{-3}$ m·K.
Ideal blackbody; T in kelvin, λ_{max} in meters. A shorter peak wavelength means higher temperature, not one exact visible color.

Worked reasoning: A thermal spectrum's peak shifts toward shorter wavelength as absolute temperature rises. Wien's relation $\lambda_{peak}\, T \approx 2.9\times10^{-3}$ m·K gives $\lambda_{peak} \approx 9.7$ micrometers at 300 K.

Your turn: Sketch two broad thermal spectra, labeling axes and the hotter peak's shift.

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two ideal thermal emitters are at 3000 and 6000 K. Which has the shorter peak wavelength? Does that mean they emit only one color?

Relative spectral emission



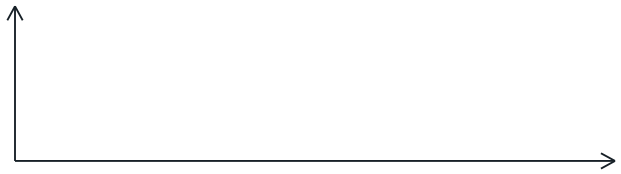
Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

04 Apply the physics

Compare ideal thermal emitters at 5000 K and 10,000 K. Use $\lambda_{peak}\, T = 2.9 \times 10^{-3}$ m·K to calculate peak wavelengths in nm. Sketch both broad spectra. Explain why their peaks alone cannot predict an exact observed meteor color.

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

An object has an infrared peak but looks red-hot. Explain how both can be true.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Across a stationary boundary, frequency remains 4 Hz while speed changes from 24 to 32 m/s. Find both wavelengths.
Label a wave sketch to support the calculation or comparison.

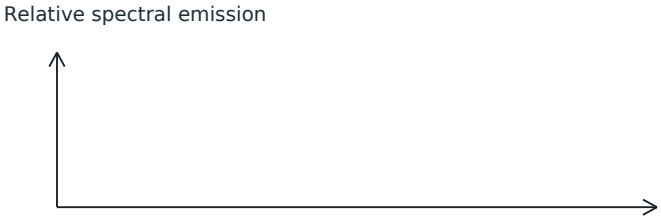


Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

An ideal blackbody has λ_{max}=725 nm. Estimate temperature.

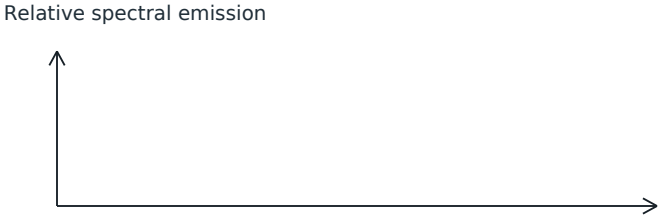


Wavelength (label units and peaks)

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Find peak wavelengths at 3000 and 6000 K. State why wavelength peak alone does not specify one exact perceived color.



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3